

Crawling gait design for a caterpillar-inspired robot with tendon-driven flexible bending segments

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Abstract—Soft crawling robots can potentially access locations that are unreachable by humans and traditional rigid robots, playing a crucial role in conducting missions like environmental monitoring or search and rescue. Due to their flexible body structures, soft robots can adapt to uncertain environments and operate safely in contact with humans. We present a new design for a caterpillar-inspired soft robot in the form of a series of 3D printed flexible tendon-driven bending segments with individual motor control. A constant-curvature quasi-static kinematic model of the robot locomotion is developed, and we describe periodic gait inputs that coordinate bending in multiple segments to lift the legs and move forward in a traveling-wave motion. We present simulations and experimental results for locomotion in a straight line, along with experimental demonstration of a modified segment design with an additional degree of freedom for steering navigation.

I. INTRODUCTION

As robotic systems attain wider adoption, the expectations of robots regarding their compatibility, adaptivity, and functionality have increased dramatically. Though traditional rigid robots have demonstrated speed and precision in highly structured manufacturing environments, new design paradigms will be necessary for robots to perform safely and robustly in unpredictable environments and unstructured interactions with humans and each other. Thanks to the development of material science and manufacturing processes, a significant amount of research has been devoted to soft robots in recent years, and the concept of soft robots seems to be a solution to the challenges robotics are now facing [1]. Soft robots have high degrees of freedom and can operate in natural environments and proximity to humans without causing damage [2]. Despite some disadvantages of soft robotics, such as difficulty to model the high-degree of freedom deformations and designing control laws, soft mechanisms still provide robots the potential to work in more complex and advanced areas such as medicine, geological exploration, and disaster rescue [3].

In soft robotics, motion is generated by various types of soft actuators. Although many soft robots use pneumatics or hydraulics to actuate their bodies [4], high fabrication quality is required to ensure air tightness. The electronic pump and valve mechanisms to control the pneumatic circuits often preclude their use in mobile robots, though recently electronics-free pneumatic oscillators have been developed and employed in a quadruped soft robot driven by compressed gas [5]. Tendon-driven actuators are often utilized

to generate bending, twisting, and contractile motion in soft robotics [6], with the length of tendons controlled by electric motors, shape-memory alloy (SMA) wires [3], or twisted-and-coiled polymers (TCA) [7].

Because animals mainly comprise soft tissues such as muscles, skins, and tendons, the natural world is one of the most important sources of inspiration for soft robotics. In the past few years, roboticists have been inspired by diverse animals in nature and have created many soft robots based on the animals' locomotion performance [8]. Among all the creatures in the natural world, caterpillars have been a rich source of inspiration for robotics researchers. Lacking any rigid skeleton, caterpillars nonetheless navigate complex 3-dimensional environments by leveraging flexible body structures that bend, extend, and contract in coordinated motion to achieve locomotion [9]. Existing work on caterpillar-inspired soft robotics includes inchworm-like motion driven by SMA coils with asymmetric frictional ground contact [3] and crawling motion driven by SMA elastomer sandwich bending actuators [10]. Motor-controlled tendon-driven caterpillar robots include designs with flexible segments that contract in length cast in Silicone [11], 3D-printed bending segments with integrated contact force sensing [12], and combination bending and contracting segments [13]. As for the material, engineers often construct their robots with elastomers to mimic the flexible body of the caterpillars. Using a casting process in 3D printed molds (as in [11]) allows for a wide variety of material choice, though the multistep fabrication process is time consuming. Direct 3D printing of soft materials like thermoplastic elastomer (TPE) offers speed and convenience, enabling complex parts with interior chambers to be produced in one step, though the materials may be less flexible and prone to fatigue. In addition, it is much easier to modify different designs without requiring redesign of both the part and the molds.

Here we present a new design for a caterpillar-inspired robot with flexible tendon-driven bending segments based on the design of Umedachi et al [12]. Our robot is designed such that the segments are bent in an arc at the neutral configuration, which allows for a greater range of motion to lift the feet off the ground than the existing design with all segments flat in the neutral stance. Additionally we introduce a second degree of freedom to one of the segments to enable steering navigation with controlled out-of-plane motion.

The main contributions of this work are to: *a*) introduce a new design for a tendon-driven modular 3D printable caterpillar-inspired soft robot with steering capabilities, *b*) demonstrate how a computationally simple quasi-static

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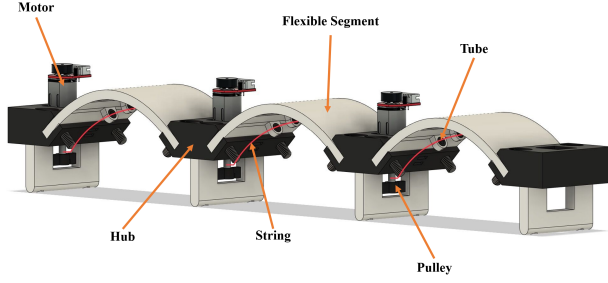


Fig. 1. CAD rendering of the caterpillar robot hardware design. Red line shows the path of the tendon string through a series of five guide tubes along the flexible segment.

constant-curvature model for planar legged locomotion can be used in the design of periodic gait control inputs, and *c*) demonstrate the crawling locomotion performance of the caterpillar-inspired soft robot that involves lifting of individual legs off the ground rather than relying completely on sliding motion, in both forwards and turning gaits.

The rest of this report is organized as follows. In section II we describe the design and construction of the robot hardware. In section III the mathematical model of the robot locomotion is described. In section IV and V we present the experimental results of the prototype robot in forward and turning motions, respectively, and in section VI we conclude with discussion of future directions for this work.

II. CATERPILLAR ROBOT HARDWARE DESIGN

We have developed a caterpillar-inspired soft robot capable of crawling locomotion on a flat surface. The robot body is composed of an alternating series of flexible tendon-driven bending segments and rigid hubs upon which the motors and prolegs are attached, as shown in Fig. 1. Rigid components including the hubs and pulleys are printed in PLA, and the prolegs and flexible segments are printed with Filaflex 60A thermoplastic elastomer filament (Recreus Industries, S.L.). The robot may be constructed with any number of segments, with the number of hubs one more than the number of segments. In the current work we consider the case of two-, three-, and four-segment caterpillars. The bending motion of each segment is actuated individually

by an N20 6V DC motor mounted in an adjacent hub. Each motor is equipped with an integrated digital encoder and a 150:1 reduction gearbox. Tendons made from nylon thread are wound up and unwound as the motor output shaft rotates, which generates the bending motion of the flexible body segments. The motors are individually controlled by an Arduino Mega microcontroller via an Adafruit Motor Shield and are powered by a 6V power supply. The controller and power supply are located offboard and are connected to the hardware prototype via flexible wire tether suspended from above. Table I summarizes the parameters of the hardware components and geometry.

A. Deformable body design

The segments are made of 3D-printed FilaFlex 60A, a highly elastic flexible filament for 3D printers. The flexible segments act as springs to store and release elastic energy and actuate the robot's motions. The flexible segments are in the shape of a rectangular prism, with five guide loops integrated along the centerline for the tendons to pass through. PLA rings are inserted into the guide loops to reduce the sliding friction of the tendon. The two ends of a segment are inserted into slots in adjacent rigid hubs and each held in place with two M3 bolts passing straight through.

B. Hub design

The 3D printed PLA hubs connect the flexible segments, the prolegs, and the motors. For the convenience of assembling and modifying the robot prototype the design uses screws and nuts to connect all the components, though it would be possible in further iterations to print the entire caterpillar in Filaflex in a single part integrating hubs, flexible segments, and prolegs. The main factor that differentiates the current design from previous tendon-driven bending caterpillar robots such as [12] is the angle at which the flexible segments are connected to the hubs. Our design uses a 45° segment connection angle, whereas [12] uses 0° such that the segments form a straight line with one another at zero curvature. Given that our bending segments have a turning angle range of motion of approximately 0 to 180° (as constrained by geometry, motor torque, and segment stiffness), the caterpillar will have a neutral stance with all feet vertical when all segments are at the midpoint of their bending range.

C. Proleg design

The prolegs are used to support the robot and exert friction when contacting the ground. The prolegs are 3D printed with FilaFlex 60A flexible filament to maximize the friction force, compared to rigid PLA material. When the proleg contacts the ground, the force produced by the robot in the horizontal direction and the friction between the legs and the ground will cause the leg to deform slightly, which further improves frictional contact. We designed and tested four different profiles for the end of the proleg (the foot) to determine the best performance for the caterpillar-inspired robot: round, rectangular, stiff triangular, and flexible triangular. The round

TABLE I
HARDWARE PROTOTYPE PARAMETER VALUES

Symbol	Name	Value
l	Flexible segment length	70 mm
d_t	Thickness of segment	4 mm
d_w	Width of segment	40 mm
w	Half-width of hub	16 mm
h	Height of hub (leg length)	32 mm
α	Segment-hub connection angle	$\pi/4$ rad
m_m	Motor mass	12 g
m_h	Hub mass (w/motor only)	24.5 g
m_h	Hub mass (w/motor and screws)	30.8 g
m_s	Flexible segment mass	16.7 g
	Encoder resolution	14 counts/rev
	Motor gear ratio	150:1 reduction

(semicircle) profile displayed the least amount of slippage during crawling motion, and it is used in all experimental results below.

III. CONSTANT-CURVATURE MODEL FOR GAIT DESIGN

We consider a planar quasi-static kinematic model of the caterpillar robot's motion under the assumption that the flexible segments connecting the hubs take on the shape of a circular arc, with the turning angles ϕ_i directly proportional to the tendon motor shaft displacement. By studying this idealized rigid system and deriving constraints on its motion, we aim to build up a methodology for gait design which can be used as a starting point for gait design in the flexible system. Geometric parameters and model variables are illustrated for a two-segment caterpillar in Fig. 2 and numerical values for the hardware prototype are summarized in Table I.

In a caterpillar with n_s flexible segments, the location of the hubs $\mathbf{r}_i = (x_i, y_i)^T \in \mathbb{R}^2$ for $i = 0, \dots, n_s$ are measured at the tip of the feet, relative to an inertial coordinate frame with origin O . The x -axis is defined as the ground, with the constraint $y_i \geq 0$ for all hubs at all times. Hub orientations θ_0 through θ_{n_s} are measured counter-clockwise from the vertical axis, such that in Fig. 2, $\theta_0 = -\pi/2$, $\theta_1 = \theta_2 = 0$.

The flexible bending segments that connect the hubs have constant length l , and take on the shape of a straight line for $\phi_i = 0$ or a circular arc with radius of curvature l/ϕ_i for $\phi_i > 0$. Flexible segments are attached to the hubs with an offset angle α , such that the orientation of successive hubs are related by

$$\theta_i - \theta_{i-1} = 2\alpha - \phi_i. \quad (1)$$

The location of a hub relative to the previous hub is described by the following in a $(i-1)^{\text{th}}$ hub-centric coordinate frame,

$$\mathbf{r}_i^{i-1} = \begin{pmatrix} w \\ h \end{pmatrix} + R(\alpha) \left[\frac{l}{\phi_i} \begin{pmatrix} \sin \phi_i \\ \cos \phi_i - 1 \end{pmatrix} + R(\alpha - \phi_i) \begin{pmatrix} w \\ -h \end{pmatrix} \right], \quad (2)$$

for $\phi_i > 0$, and

$$\mathbf{r}_i^{i-1} = \begin{pmatrix} w \\ h \end{pmatrix} + R(\alpha) \begin{pmatrix} l \\ 0 \end{pmatrix} + R(2\alpha) \begin{pmatrix} w \\ -h \end{pmatrix}, \quad (3)$$

for $\phi_i = 0$, such that in the inertial frame

$$\mathbf{r}_i = \mathbf{r}_{i-1} + R(\theta_{i-1}) \mathbf{r}_i^{i-1}, \quad (4)$$

where $R(\cdot)$ is the standard 2D rotation matrix. From (3) we derive the inter-foot distance,

$$d_i = \|\mathbf{r}_i - \mathbf{r}_{i-1}\| = 2 \left(\frac{l}{\phi_i} \sin \frac{\phi_i}{2} + w \cos \left(\alpha - \frac{\phi_i}{2} \right) + h \sin \left(\alpha - \frac{\phi_i}{2} \right) \right). \quad (5)$$

For a given caterpillar geometry, the turning angle of each segment is limited by the configuration where the two neighboring feet come in contact with each other, and we define the maximum turning angle $\bar{\phi}$ such that $d_i = 0$ for $\phi_i = \bar{\phi}$.

The full state of the caterpillar in this model is described by the vector of shape variables $\phi = (\phi_1, \dots, \phi_{n_s})^T$, along

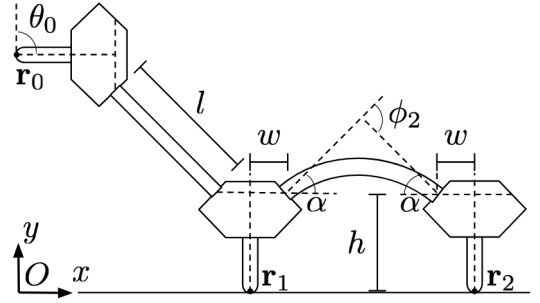


Fig. 2. Diagram of caterpillar geometry. Hub center of mass is assumed to be located where the dotted lines cross. Motors (not shown) extend from top of hub. Here $\phi_1 = 0$ (flat segment) and $\phi_2 = 2\alpha$ (parallel hubs).

with the group variables corresponding to the pose of the first hub, $(\mathbf{r}_0, \theta) \in SE(2)$ in the inertial frame. We define a “gait” $g(t)$ as a periodic trajectory in time with period T , where $\phi = g(t)$ with $g(t) = g(t+T) \forall t \geq 0$. Due to the continuous nature of the tendon-winding control of the hardware robot, we restrict our study to gaits that are continuous in time.

A. Quasi-static discrete-time kinematics model

In order to simulate the motion of the caterpillar with respect to the ground, we use the following assumptions: The center of mass of each hub is located at point $(0, h)$ in the hub-centric frame. The mass of each flexible segment is split in half, with half its mass added to mass of each neighboring hub. Gravitational force acts in the negative y direction and the line $y = 0$ represents the ground. At a given time t , two or more “stance” feet will be in contact with the ground ($y_i = 0$), and in order for a configuration to be stable, the center of mass of the entire caterpillar must lie between the left- and right-most stance foot in the x coordinate. Dry (Coulomb) friction is present at foot-ground contact points.

The simulation algorithm is as follows: From a given stable configuration $\mathbf{r}_0(t), \theta_0(t), \phi(t)$, calculate location of center of mass and determine the contact normal force for each stance foot. Whichever stance foot has the highest normal force at time t is denoted the “anchor” foot and will remain in place at the next timestep $t + \Delta t$ due to a higher friction force. Update the shape variables $\phi(t + \Delta t)$ based on the gait, and determine the group variables $\mathbf{r}_0(t + \Delta t), \theta_0(t + \Delta t), \phi(t + \Delta t)$ such that the new state is compatible with the previous stance feet. If the two stance feet from the previous step maintain stability, position the caterpillar such that the anchor foot keeps the same location. Otherwise there are two situations where the stance feet will change from one step to the next. First, if the center of mass passes over one of the stance feet, the caterpillar will roll about the anchor foot, with the anchor foot remaining at the same location. If this new state is not stable, execute additional roll motions as necessary until a stable state is found. Second, one of the previous stance feet may lift into the air if another foot reaches the ground, in which case choose the new state that keeps one of the previous stance feet in the same location. If both stance feet are lifted, position the caterpillar such that the center of mass remains in the same location. If over

an interval $[t_0, t_1]$ the stance feet remain the same and the distance between them remains constant, we denote this a “non-slip” interval.

With this algorithm, we can estimate the displacement along the x -axis over a single gait cycle to assess the performance of a gait.

B. Special classes of gaits

Below we describe a few different classes of gaits, classified by the phase relationships between segments and conditions for avoiding sliding motion of the stance feet.

1) *Traveling-wave gaits*: We denote any gait where each segment follows the same trajectory shifted in time as a *traveling-wave gait*, satisfying $\phi_j(t) = \phi_i(t + (\Delta/2\pi)(j - i)T)$, where $\Delta \in [0, 2\pi]$ is the constant phase difference between neighboring segments. If we consider hub 0 to be the rear of the robot, this type of gait with $\Delta < \pi$ mimics the rear-to-front traveling-wave crawling motion displayed by many caterpillars, such as *Manduca sexta* [9].

2) *Space-periodic gaits*: We define a space-periodic gait of period m to be a traveling-wave gait that satisfies $\text{mod}(m\Delta, 2\pi) = 0$ and

$$\sum_{i=1}^m \phi_i(t) = 2m\alpha, \forall t. \quad (6)$$

Space-periodic gaits with period m have the property that $\mathbf{r}_{i+2m} - \mathbf{r}_{i+m} = \mathbf{r}_{i+m} - \mathbf{r}_i$ at all times. Such a gait could be realized on a caterpillar robot with any number of segments, though at least $2m$ segments would be necessary for redundant foot contact. As an informal proof, consider that the relative position and orientation of one hub with respect to another depends only on the turning angles of the intermediate segments. Being m -periodic, we have $\phi_i(t) = \phi_{i+m}(t)$ for all segments i , so the sequence of segments in the chain from hub i to $i + m$ has the same bending angles as the sequence of segments in the chain from hub $i + m$ to $i + 2m$. This tells us that $\mathbf{r}_{i+2m} - \mathbf{r}_{i+m}$ has the same length as $\mathbf{r}_{i+m} - \mathbf{r}_{i+2m}$, however for the two vectors to be equal they must be parallel as well, which will be true if hub i and $i + m$ have the same orientation. By repeated application of (1) and substitution of (6), we find that $\theta_i(t) = \theta_{i+m}(t)$ under the space-periodic conditions, concluding the proof.

3) *Non-slip gaits*: A non-slip gait is a gait that satisfies the condition that within each stance interval, the stance feet maintain a constant distance between them at all times. For instance, within the class of space-periodic gaits with period $m=3$ ($\Delta=2\pi/3$), there exists a family of possible non-slip gaits parameterized by the stance phase foot distances (distance between every third hub). Figure 3 illustrates this family of gaits as closed loops in the space of bending angles, which lie on a plane $\phi_1 + \phi_2 + \phi_3 = 6\alpha$ due to the hub orientation constraint (6). Higher amplitudes of bending angles correspond to lower stance foot distances but higher displacement per cycle. For the geometric parameters used in the hardware prototype these gaits are well-approximated by sinusoid trajectories, especially at low amplitudes:

$$\phi_i(t) = 2\alpha m + A \sin(\omega t + (i - 1)\Delta). \quad (7)$$

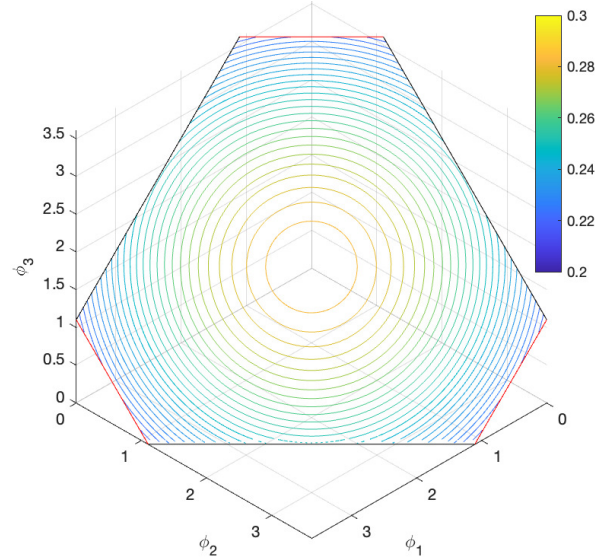


Fig. 3. Trajectory contours in shape space for non-slip space periodic gaits with $\Delta = 2\pi/3$, colored by stance foot distance in meters. The hub orientation constraint (6) places all trajectories on a 2D-plane within the 3D space of bending angles. Black and red lines represent states with minimum bending angle ($\phi_i = 0$) and maximum bending angle with feet touching ($\phi_i = \phi$), respectively.

IV. EXPERIMENTAL VALIDATION OF FORWARD GAITS

The forward crawling performance of the hardware caterpillar robot was tested under a variety of constant phase difference (traveling-wave) sinusoid gaits in the 2-, 3-, and 4-segment configurations. All combinations of two amplitudes and four phase differences were tested for each hardware configuration. Experimental trials consisted of three cycles (for 3- and 4-segment) or four cycles (for 2-segment) of the gait, and three trials were performed for each of the combinations of gait parameters and hardware configurations. Computer vision software in Matlab was used to track a set of markers placed on the hubs in order to measure the

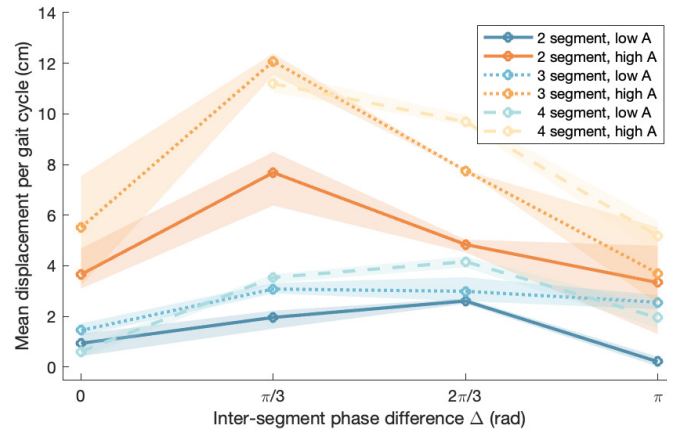


Fig. 4. Experimental results for forward displacement in traveling-wave sinusoid gaits under varying number of segments, amplitude A , and phase difference Δ . Circles represent the mean of three trials, with shading above and below corresponding to minimum and maximum individual trials for each gait combination.

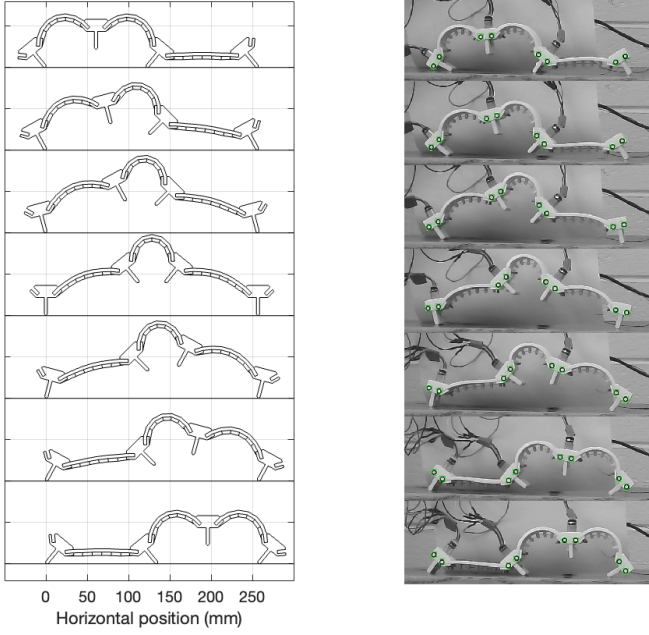


Fig. 5. Simulated (Left) and experimental (Right) snapshots of state of three-segment caterpillar during one stance interval of the $\Delta = 2\pi/3$ traveling-wave gait, with an amplitude of 450° of motor output shaft rotation. During this interval foot 0 and foot 3 remain in contact with the ground without slipping.

displacement over time from video recordings. The robot is set up so that the motor controlling a segment is to its left, leaving the rightmost hub with no motor.

To achieve the desired gait motion, a proportional control scheme was implemented in MATLAB, via serial communication with an Arduino Mega microcontroller equipped with an Adafruit Motor Shield motor controller. The motors in the caterpillar are equipped with encoders, allowing for feedback control of the individual motor shaft angles $\theta_{m,i}(t)$ through individual voltage inputs $u_i(t)$. Motor shaft angles $\theta_{m,i} = 0$ correspond to the neutral (legs parallel) configuration of $\phi_i = 2\alpha = 90^\circ$. Proportional controller gain $K_p = 6.86\text{mV}/^\circ$ was used for all experiments, with

$$\begin{aligned} u_i(t) &= K_p(\theta_{des,i}(t) - \theta_{m,i}(t)), \\ \theta_{des,i}(t) &= A \cos(\omega t + (i-1)\Delta) \end{aligned} \quad (8)$$

Two amplitudes A were used in the experiments: low amplitude of 225° pulley shaft angle corresponding to a segment bending angle range of approximately $[51^\circ, 130^\circ]$, and high amplitude of 450° corresponding to a segment bending angle range of approximately $[11^\circ, 169^\circ]$.

The walking gait displacement experimental results over all trials are summarized in Fig. 4. The data suggests a trend of increased displacement from both increasing the number of segments and increasing the amplitude, with maximum average displacement per cycle observed for the 3-segment robot with high amplitude and $\Delta = \pi/3$. More detailed results for a single 3-segment trial with high amplitude and $\Delta = 2\pi/3$ are illustrated as snapshots in Fig. 5 and trajectory plots in Fig. 6. This particular gait approximates a non-slip

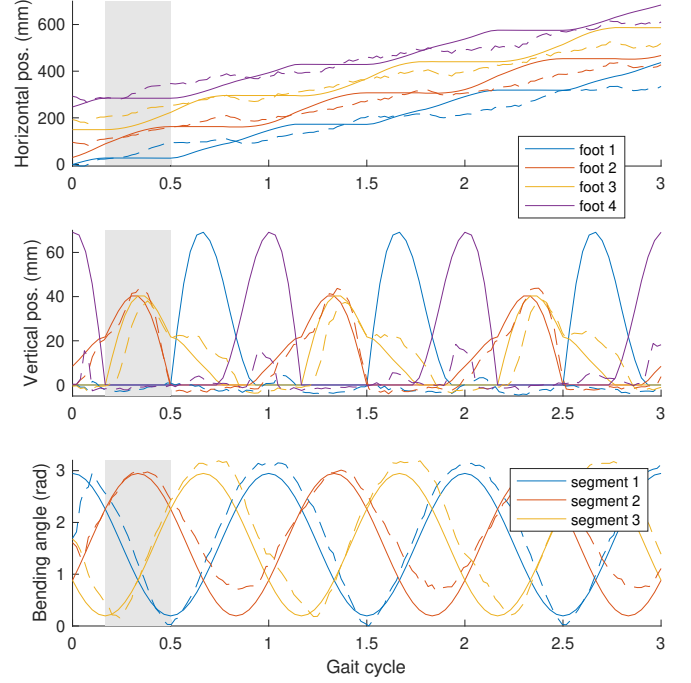


Fig. 6. Simulated (solid) and experimental (dashed) trajectories of 3-segment caterpillar over three cycles of the $\Delta = 2\pi/3$ traveling-wave gait, with an amplitude of 450° of motor output shaft rotation. First and second plot show the horizontal and vertical displacement of the feet, and the third plot shows the flexible segment turning angles ϕ_i . Shaded region represents the time interval illustrated in Fig. 5 above.

gait during the portion of the cycle where the stance feet are on the first and last hub. The experiment matches well with the simulation result, especially during the non-slip interval when each segment is supported on both sides.

V. EXPERIMENTS WITH STEERING CONTROL

In order to enable steering navigation for the robot, we developed a modified Filaflex bending segment with an additional motor mounted in the adjacent hub for out-of-plane bending. In this case, the deformable body is designed to have a “fishbone” shape, which has a backbone in the middle and several branches at each side of the backbone. Each branch has a hole in it to let the tendon pass through, with a PLA ring inserted to reduce friction on the tendon. To ensure the backbone can provide sufficient restoring force, the second moment of inertia of the backbone’s square cross-section is designed to be the same as the standard flexible bending segment described above. There are two tendons for the steering motion, one on either side, which are wound around the steering motor shaft in opposite directions to facilitate both positive and negative steering curvature.

Replacing the center segment of the 3-segment prototype with the modified steering segment, we tested the steering ability of the robot. In each trial, the steering motor angle is held constant, while the bending segment motors perform the $\Delta = 2\pi/3$ traveling-wave gait with amplitude 400° over five cycles. The turning motion of the prototype is measured by the angle, Ψ , of the line connecting the first and last motors, relative to the starting orientation. We observe a linear

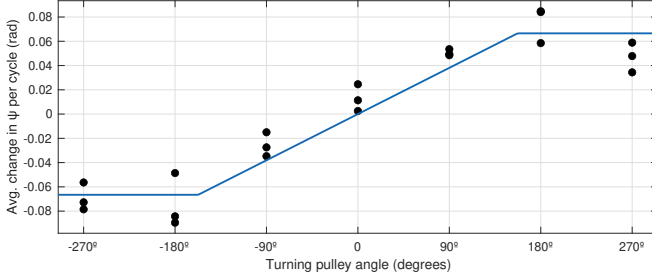
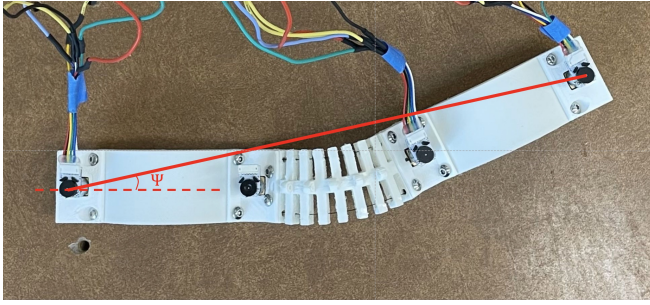


Fig. 7. Steering performance of the modified caterpillar robot. Top: orientation of robot is determined by line passing through first and last hub motors in overhead view. Bottom: Average turning rate Ψ /cycle. Black circles are experimental data from three trials of five cycles each for each turning motor angle. Line is best fit of a linear function with saturation.

relationship between steering motor angle and turning rate at low angles, which saturates at higher angles, illustrated in Fig. 7. The best fit line of this form has a slope of 4.23×10^{-4} rad per cycle turning rate per degree of steering motor angle, with saturation at 0.0665 rad, via MATLAB Curve Fitting Toolbox. At higher steering angles than 270° the robot is at risk of falling over. Additional experiments were conducted with the steering segment at the front and rear, but it was found that placing the steering segment in the center provided the best turning rate and was more resistant to falling over during locomotion.

VI. DISCUSSION AND FUTURE DIRECTIONS

This research presents a caterpillar-inspired soft robot capable of crawling gaits on the horizontal surface with forward and steered motion. We demonstrate that a simple constant-curvature model of the robot motion provides sufficient simulation accuracy for the purposes of gait design for the hardware prototype. Careful design of periodic gait inputs enable the robot to lift its legs during crawling, so as to not rely totally on frictional sliding. The robot's structural components are fabricated by 3D printing, making the manufacturing process accessible, easy to modify, and low cost.

In future work, a motion model that incorporates the flexibility of the segments may provide improved accuracy for simulation and gait design. While simulation efforts thus far have focused on the particular hardware prototype, optimization may be performed both on the gait trajectories as well as on the geometric parameters of the caterpillar hardware design to maximize displacement or efficiency of locomotion. The current hardware relies on a tether wire

for control and power delivery, which while convenient for testing limits its utility. Future revisions could incorporate controllers and batteries by employing flexible circuit boards, as in [10]. The optimal number of segments may be assessed as well, which may present a trade-off between increased locomotion performance, load capacity, and power consumption as additional motors are added. Lastly we consider that the addition of active gripping capabilities to the tips of the prolegs would enable locomotion beyond horizontal ground to climb tree branches and other natural environments.

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